

Diffusion and Flow-Based Models for Controllable Image Generation

Abstract

Controllable image generation has evolved from a secondary design goal into a central criterion for modern generative modeling. The recent dominance of diffusion models in text-to-image synthesis has been matched by a rapid proliferation of control mechanisms that target spatial grounding, compositionality, user instruction following, geometric consistency, identity preservation, personalization, and real-image editing. In parallel, flow-based formulations—especially flow matching and rectified flow—have emerged as a closely related, often computationally favorable alternative that increasingly powers state-of-the-art image generators and editors. This survey synthesizes these two lines under a unified perspective: controllability can be understood through the interaction between the control signal, the intervention locus within the generator, and the optimization regime at which control is imposed. On the diffusion side, the field now spans guidance-based methods, architectural adapters, grounded generation modules, inversion-based editors, and multimodal personalization systems. On the flow side, recent work shows that controllable generation can be realized through vector-field steering, inversion-free transport between source and target distributions, optimal-control formulations, and hybrid diffusion-to-flow transfer. We argue that diffusion and flow models should not be treated as isolated families but as overlapping points in a broader continuous-time generative design space. We review the mathematical foundations, organize the major method families, highlight empirical evaluation practices, and analyze persistent challenges, including the controllability-fidelity trade-off, compositional failures, benchmark fragility, person-specific identity drift, efficient real-image editing, and safety under increasingly capable open models. The survey concludes that the most promising future directions lie in unified multimodal control interfaces, training objectives that explicitly reward faithful adherence to structured constraints, and evaluation protocols that measure control at the instance, relation, and trajectory levels rather than only through global perceptual quality. ¹

Introduction

The modern controllable image generation literature is built on two closely connected developments. The first is the rise of diffusion and score-based models, beginning with nonequilibrium diffusion formulations and culminating in denoising diffusion probabilistic models, score-based stochastic differential equation methods, improved samplers, distillation, and latent-space text-to-image systems. The second is the shift from pure text conditioning toward rich external control, motivated by the observation that unconstrained prompt-based synthesis remains unreliable for object counting, attribute binding, scene layout, geometric editing, and identity preservation. Recent surveys therefore converge on a common diagnosis: image quality alone is no longer a sufficient metric of progress; utility depends on whether a model obeys user intent at fine granularity. ²

Within this literature, “controllability” is not a single capability but a bundle of related requirements. A controllable generator may need to respect structural conditions such as edges, depth maps, human pose,

segmentation masks, keypoints, or bounding boxes; semantic conditions such as object attributes, counts, relations, or subprompts; reference conditions such as exemplar images, identities, styles, or user-specific concepts; or optimization-oriented conditions such as rewards, reconstruction losses, physics constraints, or editing instructions. One of the clearest conceptual contributions of the recent survey literature is to separate the nature of the condition from the mechanism of control. The same pose signal, for example, may be injected through an auxiliary encoder, copied into a frozen backbone via ControlNet-style branches, imposed by attention modulation at inference time, or enforced by solving an optimization problem over the sampling trajectory. ³

A second conceptual shift is that diffusion and flow models increasingly occupy a shared research space rather than two disjoint camps. Flow matching was introduced as a simulation-free way to train continuous normalizing flows through regression on vector fields along predefined probability paths, while rectified flow emphasized learning straighter transport trajectories to enable fewer function evaluations. Subsequent image-generation systems scaled these ideas to transformer backbones and high-resolution text-to-image synthesis, while later work connected flow and diffusion parameterizations more explicitly and exploited their practical compatibility for transfer and hybridization. In this sense, the current generation of controllable models is best understood as operating in a continuous-time generative family with multiple equivalent or near-equivalent parameterizations, rather than through a hard boundary between “diffusion” and “flow.” ⁴

This survey adopts that unified perspective and focuses specifically on controllable image generation. It does not attempt to cover the entire diffusion literature, nor does it treat video or 3D generation as primary topics except where image-specific methods descend directly from those settings. The principal thesis is that controllability research has advanced along three orthogonal axes: the representation of conditions, the sites at which those conditions enter the model, and the degree to which control is learned offline versus imposed online. Under this view, classical classifier-free guidance, ControlNet-style branching, real-image inversion, localized attention editing, exemplar-based personalization, multi-instance layout synthesis, and rectified-flow vector-field steering can be read as different answers to the same overarching design question: how should a generative trajectory be constrained so that it remains both faithful to the user and plausible under the image prior? ⁵

Methods

Diffusion-centered control mechanisms

A useful starting point is the core diffusion formulation. In the standard denoising framework, image generation proceeds by reversing a forward noising process, either as a discrete Markov chain or its continuous-time counterpart. Control enters this process in three broad ways. The first is **guidance**, where gradients, scores, or surrogate objectives alter the sampling trajectory without changing the generator weights. Classifier guidance and classifier-free guidance established the basic recipe, but later work generalized it substantially: universal guidance, loss-guided plug-and-play control, diffusion self-guidance, diffusion posterior sampling, and related methods show that pre-trained generators can be repurposed for inverse problems, reward maximization, semantic editing, or constrained synthesis by acting directly on the denoising dynamics. The main attraction of this family is modularity—one can often reuse a frozen model—but the main limitation is instability when the desired constraint conflicts strongly with the native image prior or must be enforced locally rather than globally [1-18,33-38]. ⁶

A second family modifies the **conditioning architecture** itself. Latent diffusion and its descendants made the text-conditioned setting computationally practical, but they also exposed multiple insertion points for external control: new channels, side encoders, residual branches, gated attention modules, or token-level augmentations. ControlNet demonstrated that a powerful frozen backbone can be copied into a trainable branch and safely adapted to conditions such as canny edges, depth, pose, and segmentation. T2I-Adapter pursued a lighter-weight variant through learned adapters. GLIGEN extended pre-trained diffusion models to open-set grounded generation with textual phrases and bounding boxes. Uni-ControlNet, Composer, and AnyControl addressed mixed and composable conditions, while later work introduced harder region binding, local-only control, and multi-instance controllers such as InstanceDiffusion, MIGC, MIGC++, and 3DIS. Collectively, these methods move the field from “text plus optional hint” toward a richer view in which image generation becomes a structured rendering process conditioned on instance-level or region-level constraints [19-33,51-74]. ⁷

A third and especially influential family addresses **editing and inversion**. Here the goal is not merely to synthesize a new image from conditions, but to transform a specific source image while preserving non-target content. SDEdit and RePaint illustrated early diffusion-based editing by partial noising and conditional denoising. Prompt-to-Prompt identified cross-attention as a key locus for controlling semantic edits through prompt-level manipulations. Null-text inversion, EDICT, negative-prompt inversion, direct inversion, and task-oriented inversion improved the mapping from real images into editable latent noise, while Pix2Pix-Zero, Plug-and-Play Diffusion Features, MasaCtrl, LEDITS++, DragDiffusion, DragonDiffusion, Diffusion Handles, SINE, and TurboEdit broadened the edit space to include zero-shot translation, non-rigid editing, drag-based manipulation, 3D-aware transformations, and fast instruction-guided modification. The common design tension is that stronger preservation of source structure often reduces edit strength, while aggressive manipulation typically amplifies artifacts or destroys identity [39-50,75-99]. ⁸

A closely related subfield, but important enough to deserve separate analytical emphasis, is **personalization and reference control**. Textual Inversion, DreamBooth, and Custom Diffusion established the modern subject-driven paradigm by associating user concepts with learnable embeddings or lightweight fine-tuning. Subsequent work split roughly into two directions. One direction improved training-based customization through tuning encoders, multi-concept fusion, region-aware merging, and prompt-aligned personalization. The other direction minimized or eliminated test-time optimization through exemplar encoders and adapter schemes, including FastComposer, IP-Adapter, ELITE, BLIP-Diffusion, Subject-Diffusion, PhotoMaker, InstantID, and PuLID. These methods reveal that controllability in the personalized regime is not just a question of preserving appearance; it also includes editability under new prompts, multi-subject disentanglement, cross-image consistency, and the ability to retain non-identity attributes such as pose, clothing, background, or style. In practice, personalization has become one of the clearest demonstrations that “conditioning” is inseparable from evaluation: a method that perfectly reconstructs the reference but fails to obey the prompt is not controllable in the intended sense [100-118]. ⁹

Flow-centered and hybrid control mechanisms

Flow matching and rectified flow start from a different training view but lead to remarkably similar control questions. Flow matching learns a time-dependent vector field that transports a simple source distribution to the data distribution through regression on analytically defined conditional paths. Rectified flow further privileges straight trajectories, often with the explicit goal of reducing sampling curvature and thus the number of numerical solver steps. Recent large-scale image generators—including transformer-based

models trained with rectified-flow or flow-matching objectives—show that these formulations can reach or exceed diffusion-era image quality while benefiting from cleaner continuous-time dynamics and fewer function evaluations. This efficiency is not merely an engineering convenience. It qualitatively changes controllability, because test-time control methods become more attractive when the underlying trajectory is shorter, more stable, and easier to manipulate mathematically [16-18,30-31,119-124]. ¹⁰

Recent controllable flow methods can be grouped into four themes. The first is **training-free guidance of the vector field**. FlowChef steers rectified-flow models directly in vector-field space for classifier-guided generation, inverse problems, and editing, emphasizing inversion-free and low-memory control. OC-Flow formulates guided flow matching through optimal control, placing earlier backprop-through-ODE approaches inside a more principled framework. Source-Guided Flow Matching shifts the intervention from the learned field to the source distribution itself, offering a conceptually distinct route to exact guidance. These methods are important because they suggest that control in flow models may be more naturally phrased as optimal transport or trajectory design, not merely as score perturbation [121,126,127,136]. ¹¹

The second theme is **flow-native editing and inversion**. Stable Flow adapts attention injection to rectified-flow transformers by identifying “vital layers” that support consistent edits. RF-Solver and related work reduce inversion error by improving the numerical solution of rectified-flow ODEs, enabling stronger image and video editing. Semantic Image Inversion and Editing using Rectified Stochastic Differential Equations shows that rectified-flow inversion can also be derived from control-theoretic arguments. FluxSpace and latent-space editing methods exploit the internal representations of flow transformers to produce disentangled semantic edits, while FlowEdit replaces editing-by-inversion with an explicit transport ODE from source prompt distribution to target prompt distribution. BiFM extends this line further by jointly learning generation and inversion in a bidirectional few-step framework. Compared with diffusion editing, the major unresolved issue is not whether flow models can edit, but which editing paradigm—exact inversion, stochastic inversion, vector-field steering, or direct transport—best balances fidelity, flexibility, and runtime [125,128-133]. ¹²

The third theme is **personalization and alignment in flow systems**. RectifID adapts anchored classifier guidance to reference-driven rectified-flow personalization. RFMI proposes mutual-information-based self-supervision for improving text-image alignment in rectified-flow text-to-image models without external reward models. FLUX.1 Kontext goes even further by unifying generation and editing in a single latent-space flow-matching architecture with text and image context tokens, blurring the classic distinction between “generation model” and “editor.” This signals a broader shift: as flow backbones become foundation models rather than narrow samplers, control increasingly migrates from specialized external modules into the native interface of the generator itself [31,121,124,134]. ¹³

The fourth theme is **diffusion-flow hybridization**. SiT already suggested that diffusion and flow-style objectives can be studied within a shared interpolant framework. Rectified Diffusion argued that what matters operationally is not literal straightness but the learned approximation of an efficient ODE path. Diff2Flow systematized transfer from pre-trained diffusion models to flow matching, while more recent alignment work makes explicit that many practical tricks—parameterization choices, timesteps, schedules, guidance heuristics, and initialization schemes—can be translated across the diffusion-flow boundary. For controllable generation, this hybrid view is especially fertile. It implies that architectural advances from diffusion, control modules from grounded generation, and trajectory manipulation techniques from flow matching need not remain siloed. The most promising systems are likely to be hybrids: diffusion-derived

backbones or adapters trained or fine-tuned in flow form, coupled with control objectives that exploit the mathematical clarity of continuous transport [16-18,119,122-136,149-150]. ¹⁴

Challenges and Prospects

The empirical literature now makes clear that controllability is not a monotone function of model scale. Larger text-to-image models often improve global plausibility, but compositional control remains brittle. Benchmarks such as GenEval, T2I-CompBench, VISOR, DALL-Eval, HRS-Bench, DreamBench++, RefVNL, DSH-Bench, and more recent reasoning-aware evaluations repeatedly show failures in counting, relation grounding, attribute binding, long-context instruction following, and subject preservation. The central methodological problem is that most current control methods optimize local adherence to a subset of conditions while assuming that the remaining image prior will “fill in the rest.” This assumption fails for entangled scenes, multi-subject prompts, or edits that require coordinated global and local changes simultaneously [137-150]. ¹⁵

A second challenge is the **controllability-fidelity trade-off**. Guidance-time methods can overconstrain the trajectory and induce oversaturation, repetition, or structural collapse. Adapter-based methods may improve structure but weaken text alignment when conditions conflict with the prompt. Inversion-based editors preserve real-image content but often narrow the reachable edit space. Personalized systems preserve identity at the cost of overfitting and prompt under-responsiveness. Flow systems inherit the same conflict under different language: steering the vector field or transport path can preserve source semantics too aggressively, or conversely can induce transport trajectories that satisfy the target condition while drifting from the source manifold. Future progress therefore requires objectives and interfaces that explicitly expose trade-off knobs, rather than burying them in opaque hyperparameters such as guidance scales, edit strengths, or layer injection schedules. ¹⁶

A third challenge concerns **evaluation itself**. FID remains useful for unconditional quality, but it is fundamentally misaligned with controllable synthesis because it ignores whether a specific condition was met. CLIP-based scores are more condition-aware, but they are coarse and often insensitive to instance-level failures. Object-focused metrics help with counting and layout, while identity metrics help with personalization, and edit benchmarks better capture preservation. Yet no single metric captures the full problem. The strongest recent trend is therefore metric pluralism: control should be evaluated simultaneously at the levels of prompt faithfulness, instance binding, spatial relation accuracy, source preservation, identity consistency, and human preference. A longer-term opportunity is evaluator models trained specifically for control, not just generic image-text alignment. ¹⁷

A fourth challenge is **efficiency under structured control**. Modern systems increasingly combine multiple conditions—text, layout, reference imagery, masks, style exemplars, and edit instructions—in a single workflow. But each added channel increases both computational and interface complexity. The success of rectified-flow and flow-matching systems suggests one path forward: if the base trajectory can be shortened, more of the computational budget can be spent on control rather than on denoising itself. Another path is more modular control composition, where models learn a shared control representation that can encode heterogeneous condition types instead of requiring one branch per modality. A third, and perhaps deepest, path is data-centric: training corpora need richer annotations for relations, edits, and identities, otherwise control methods remain forced to improvise supervision with detectors, pseudo-labelers, or synthetic decomposition pipelines. ¹⁸

Finally, the field faces nontrivial **safety and governance** questions. Precise controllability is not only a creative affordance; it also lowers the technical barrier for impersonation, document falsification, deceptive editing, and targeted misinformation. Identity-preserving personalization, real-image editing, and grounded scene manipulation are precisely the capabilities that make misuse easier. Technical research should therefore treat watermarking, provenance, edit detectability, identity-use restrictions, and auditable control logs as first-class problems rather than downstream deployment concerns. Methodologically, this suggests an additional design criterion for future controllable generators: not only how well a model can follow a condition, but how transparently it can report what conditioning sources were used and which regions or concepts were altered. 19

Conclusion

Diffusion and flow-based models have transformed controllable image generation from a collection of specialized conditional pipelines into a broad, mathematically connected design space. Diffusion models supplied the initial breakthroughs in guidance, inversion, grounded conditioning, and personalization. Flow matching and rectified flow subsequently demonstrated that many of these control problems can be reformulated in terms of vector-field steering, path design, and transport between conditioned distributions. The clearest lesson of the literature is that control is not a single module to be attached to a generator after the fact. It is an architectural, algorithmic, and evaluative property that emerges from how conditions are represented, where they are injected, and how trajectories are optimized. The next stage of progress will likely come from unified multimodal interfaces, hybrid diffusion-flow training schemes, stronger compositional supervision, and evaluation protocols that treat control as a structured prediction problem rather than a side effect of perceptual realism. 20

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